INCH-POUND

A-A-52102
April 15, 1991

COMMERCIAL ITEM DESCRIPTION

DETERGENT, LAUNDRY (NON-PHOSPHATE, LOW TEMPERATURE)

The General Services Administration has authorized the use of this commercial item description.

Abstract. This commercial item description covers a detergent for laundering cotton, cotton/polyester blend, and synthetic fabrics.

1. SALIENT CHARACTERISTICS

1.1 Description. This commercial item description covers low temperature, non-phosphate laundry detergent. The detergent shall contain softening and anti-static/anti-redeposition properties. The detergent shall not contain any optical brighteners, bluing agents, or bleach.

1.2 Appearance. The laundry detergent shall be granular and non-dusting, uniform in appearance, and free of caking or lumping. Any caking or lumping which occurs during normal storage shall be easily broken up by hand.

1.3 Chemical and physical requirements. The non-phosphate, low temperature detergent (90 to 100°F) shall conform to the requirements shown in table I when tested in accordance with the specified test methods.

TABLE I. Requirements

Requirements	Percent Min.	Percent Max.	Test methods ASTM
Free alkalinity, as Na_2O	20.0	25.0	D 501
Carbonates, as Na_2CO_3	40.0	45.0	D 501
Silicates, as SiO_2	6.0	10.0	D 460

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to: Commander, U.S. Army Natick Research, Development, and Engineering Center, ATTN: STRNC-IRT, Natick, MA 01760-5019.

AMSC N/A FSC 7930

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TABLE I. Requirements (cont'd)

Requirements	Percent Min.	Percent Max.	Test methods ASTM
Surfactant, synthetic organic	5.0	--	

1.4 Color. The color of the laundry detergent shall be white. Presence of dark or foreign particles is unacceptable.

1.5 Odor. The laundry detergent shall not have an odor of solvent or pungent, acrid or other unpleasant odor, and shall not impart such odors to fabrics washed in the detergent.

1.6 Solubility. The laundry detergent shall be totally dissolved when tested as follows:

To a 2 grams sample of detergent, add sufficient distilled water to make 100 mL of solution. The solution shall be maintained at room temperature ($75^{\circ} \pm 2^{\circ}F$) and stirred vigorously for 5 minutes. The solution shall be inspected immediately for undissolved material. Presence of undissolved material shall constitute failure of this test.

1.7 pH. The laundry detergent shall have a minimum pH of 11.0 and a maximum pH of 11.5 when tested in accordance with the following precedure:

Prepare a 1.0 percent solution of the detergent in freshly boiled, distilled, or deionized water which has been cooled to room temperature ($77^{\circ} \pm 2^{\circ}F$). Measure the pH of the detergent solution at room temperature with a pH meter having a sensitivity and readability of 0.05 pH units and a glass electrode that has been standardized with a buffer solution of pH 10.0. Report to the nearest 0.1 pH and do not correct the results for sodium ion concentration.

2. CERTIFICATION

2.1 Certification. The contractor shall certify, and maintain substantiating evidence, that the product offered meets the salient characteristics and requirements of this Commercial Item Description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

2.2 Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practicable.

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3. PACKAGING

3.1 Packaging. Forty-nine ounces of detergent shall be packaged in a paperboard box. Twenty-five pounds shall be packaged in a fiber drum. Fifty-five pounds shall be packaged in a multiwall bag.

3.2 Packing. Twelve 49-ounce boxes shall be packed in a fiberboard box conforming to RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Box shall be closed in accordance with the appendix of PPP-B-636. Drums and bags need not additional packing.

3.3 Palletization. When specified (see 4.1), detergent, packed as specified, shall be palletized in accordance with load type Ia, XIII or XV, as applicable, of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV, or type V of MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C and D or film bonding means F and G. Pallet pattern shall be in accordance with the appendix of MIL-STD-147.

3.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers and unit loads shall be marked in accordance with MIL-STD-129.

4. NOTES

4.1 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this document.
b. Size container (package) required (see 3.1).
c. When palletization is required (see 3.3).

4.2 Source of non-government documents.

ASTM D 460 - Sampling and Chemical Analysis of Soaps and Soaps Products
ASTM D 501 - Sampling and Chemical Analysis of Alkaline Detergents

Available from:

The American Society for Testing and Materials (ASTM)
1916 Race Street
Philadelphia, PA 19103-1187

4.3 Source of government documents. Copies of Military and Federal documents are available from:

Standardization Documents Order Desk
Building 4D
700 Robbins Avenue
Philadelphia, PA 19111-5094

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